

Tuesday, March 31

- James Enouen (USC)
  - ↳ Estimating Shapley values (my research interest)
  - ↳ Thursday 12pm in Roberts North 15

Today

Suyog Soti (Databricks) !!  
😊

↳ Fairview → CU Boulder → Google → Databricks

↳ real load balancing

## Power Method

Motivation: Compute  $\lambda_k$  in  $\tilde{O}(ndk)$  time instead of  $O(nd^2)$  time

$$X^T X = \sum_{i=1}^d v_i v_i^T \sigma_i^2 = A = \sum_{i=1}^d v_i v_i^T \lambda_i$$

Goal: Compute  $z$  s.t.  $\|z - v_i\|_2 \leq \epsilon$  in  $O\left(nd \frac{\log d/\epsilon}{\gamma}\right)$  time

$$z^{(0)} \sim \mathcal{N}(0, I_d)$$

$$z^{(0)} = c_1^{(0)} v_1 + c_2^{(0)} v_2 + \dots + c_d^{(0)} v_d$$

since  $v_1, \dots, v_d$  is a basis

$$\eta^{(t)} = \|z^{(t)}\|_2$$

$$z^{(t+1)} = A z^{(t)} / \eta^{(t)}$$

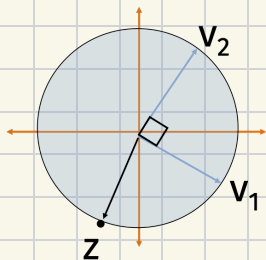
Return  $z^{(q)}$

where

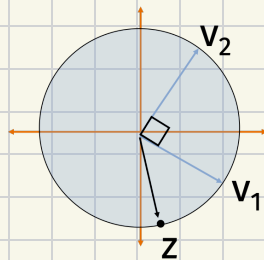
$$q = \frac{\log(d^{3.5}/\epsilon)}{\gamma}$$

# Geometric Intuition

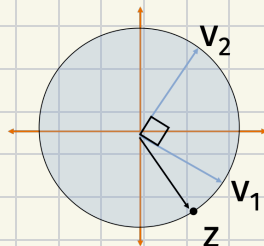
0 iterations



1 iterations



2 iterations



$$z^{(g)} = \frac{1}{\pi_{\eta}(t)} A^g z^{(0)} =$$

$$z^{(q)} = \frac{1}{\prod_t \eta^{(t)}} A^q z^{(0)}$$

Fact:  $\left| \frac{c_j^{(0)}}{c_1^{(0)}} \right| \leq d^3$  whp

Fact:  $(1-\gamma)^{1/\gamma} \leq 1/e$

$$\gamma = \frac{\lambda_1 - \lambda_2}{\lambda_1} \leq \frac{\lambda_1 - \lambda_j}{\lambda_1}$$

$$\left| \frac{c_j^{(q)}}{c_1^{(q)}} \right| = \left| \frac{c_j^{(0)}}{c_1^{(0)}} \right| \left( \frac{\lambda_j}{\lambda_1} \right)^q \leq d^3 \left( 1 - \frac{\lambda_1 - \lambda_j}{\lambda_1} \right)^q \leq d^3 \left( (1-\gamma)^{1/\gamma} \right)^{\log(d^3 \sqrt{d/\epsilon})} \leq \sqrt{\epsilon/d}$$

Since  $\sum_{i=1}^d (c_i^{(q)})^2 = 1 = (c_1^{(q)})^2 \left( 1 + \sum_{i=2}^d \left| \frac{c_i^{(q)}}{c_1^{(q)}} \right|^2 \right) \leq (c_1^{(q)})^2 (1 + \epsilon/2)$

$$\Rightarrow \left| c_1^{(q)} \right| \geq 1 - \epsilon$$

↑  
hiding  
some constants

then  $\|z^{(q)} - v_1\|_2^2 = 2 - 2 \langle z^{(q)}, v_1 \rangle \stackrel{\text{wlog}}{\leq} 2 - 2(1-\epsilon) \leq 2\epsilon$